Shuffle experiment summary from invariance CV

Mean coefficient of variation from io_together/full_shuf_v1; each block is compared to its own experiment baseline.

Ablation	Result on each metric (mean CV)	Inc/dec/same vs block baseline	Interpretation
C. elegans (real wt.)	MC 0.306 | IPC 0.212 KR 0.264 | GR 0.442	real-wt baseline	Real-weight reference for degree-preserving shuffle controls.
CE weight shuffle	MC 0.342 | IPC 0.251 KR 0.353 | GR 0.557	MC inc +12%; IPC inc +18% KR inc +34%; GR inc +26%	Real-weight shuffles all move by nearly the same amount; disrupting heterogeneous weight placement lowers invariance.
Connection shuffle only	MC 0.339 | IPC 0.249 KR 0.356 | GR 0.563	MC inc +11%; IPC inc +18% KR inc +35%; GR inc +27%	Matches the weight-shuffle effect, implying rewiring mostly acts by moving biological weights to new edges.
Connection + weight shuffle	MC 0.338 | IPC 0.254 KR 0.355 | GR 0.564	MC inc +10%; IPC inc +20% KR inc +35%; GR inc +28%	Key confirmation: adding weight shuffle after connection shuffle does not add a new effect; connection shuffle's effect is weight reassignment.
+/-1 sign-pres. baseline	MC 0.284 | IPC 0.212 KR 0.261 | GR 0.438	+/-1 baseline	Magnitude heterogeneity is removed while preserving the original sign assignment.
+/-1 sign shuffle	MC 0.277 | IPC 0.212 KR 0.266 | GR 0.447	MC same -3%; IPC same +0% KR same +2%; GR same +2%	Preserving only global E/I balance is close to local +/-1; exact edge-by-edge sign placement is less important for IPC/KR/GR.
+/-1 connection shuffle	MC 0.269 | IPC 0.216 KR 0.264 | GR 0.452	MC dec -5%; IPC same +2% KR same +1%; GR same +3%	After weights are collapsed to +/-1, GR/KR/IPC are largely insensitive to rewiring; MC carries residual topology sensitivity.
+/-1 conn. + sign shuffle	MC 0.266 | IPC 0.214 KR 0.260 | GR 0.446	MC dec -6%; IPC same +1% KR same -0%; GR same +2%	Adding sign shuffle after +/-1 rewiring changes little for GR/KR/IPC; MC remains the main topology-sensitive metric.
Unsigned binary baseline	MC 0.275 | IPC 0.209 KR 0.247 | GR 0.417	binary baseline	All nonzero edges have identical sign and magnitude, isolating topology most directly.
Unsigned binary topology shuffle	MC 0.247 | IPC 0.206 KR 0.238 | GR 0.414	MC dec -10%; IPC same -1% KR same -3%; GR same -1%	In the unsigned binary control, IPC/KR/GR remain stable, while MC is the metric most sensitive to topology.

Higher CV = less invariant; lower CV = more invariant. Inc/dec/same uses a +/-5% mean-CV threshold.